Session 16

Dart Maps Assignment

Objective: Learn how to create, access, update, and manipulate Maps in Dart using practical examples.

Task 1: Create a Map

Description: Create a Map named **foodItem** with keys **name**, **price**, and **category**, then print each value.

```
void main() {
  Map<String, dynamic> foodItem = {
    'name': 'Paneer Pizza',
    'price': 299,
    'category': 'Fast Food'
  };

  print(foodItem['name']);
  print(foodItem['price']);
  print(foodItem['category']);
}
```

Expected Output:
Paneer Pizza
299
Fast Food

Task 2: Update a Map Value

Description: Update the **inStock** key in a Flipkart product Map to **false**.

```
Map<String, dynamic> product = {
  'title': 'Wireless Mouse',
  'price': 799,
  'inStock': true
};

product['inStock'] = false;
print(product);
```

Expected Output:
{title: Wireless Mouse, price: 799, inStock: false}

Task 3: User Display Function

Description: Create a function that returns the user's display name in the format **lastName, firstName**.

```
String getUserDisplayName(Map user) {
  return '${user['lastName']}, ${user['firstName']}';
}

void main() {
  Map user = {
    'firstName': 'Jainish',
    'lastName': 'Patel'
  };

  print(getUserDisplayName(user));
}
```

Expected Output:
Patel, Jainish

Task 4: Update Playlist

Description: Add a new song to the **songs** list inside a Spotify playlist Map.

```
Map<String, dynamic> playlist = {
  'playlistName': 'Chill Vibes',
  'songs': ['Song A', 'Song B', 'Song C']
};

playlist['songs'].add('Song D');

print(playlist);
```

Expected Output:
{playlistName: Chill Vibes, songs: [Song A, Song B, Song C, Song D]}

Task 5: Refactor Using Map

Description: Store restaurant information in a Map and print all details using string interpolation.

```
void main() {
  Map<String, dynamic> restaurant = {
    'name': 'Delhi Express',
    'rating': 4.5,
    'cuisine': 'North Indian'
  };

  print(
    '${restaurant['name']} | Rating: ${restaurant['rating']} | Cuisine: $
{restaurant['cuisine']}'
  );
}
```

Expected Output:
Delhi Express | Rating: 4.5 | Cuisine: North Indian

Conclusion: This session covered Dart Maps, including creating key-value pairs, accessing values, updating data, using nested collections, and passing Maps to functions.